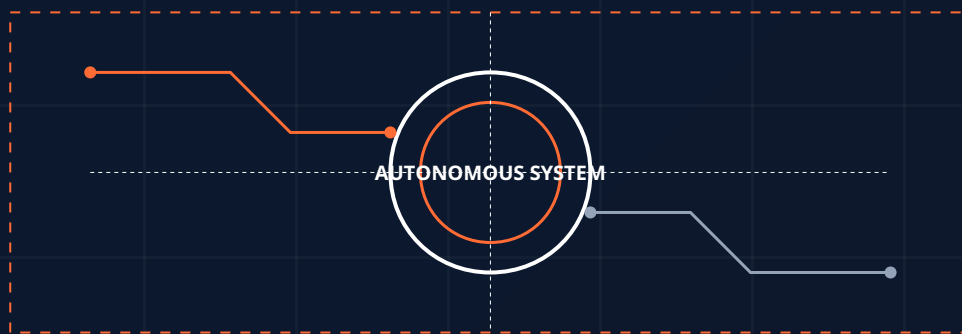


ENGINEERING PORTFOLIO

MICROMOUSE ROBOT

AUTONOMOUS MAZE SOLVING ROBOT



EVENT

Techfest IIT Bombay

Asia's Largest Science & Technology Festival

DEVELOPED BY

Vikas

Robotics & Embedded Systems Portfolio

PROJECT OVERVIEW

UNDERSTANDING THE AUTONOMOUS CHALLENGE

What is Micromouse?

Micromouse is an engineering event where small autonomous vehicles solve an unknown 16x16 maze grid. The robot must self-navigate from a starting corner to the center destination without remote control, external communications, or prior layout knowledge.

Purpose of the Project

The primary goal was to construct a robust hardware platform combined with algorithmic maze-mapping logic capable of real-time decision making under physical sensor constraints and strict dynamic limits.

Core Metrics

Maze Grid: 16 x 16 Cells

Cell Width: 18 cm

Wall Height: 5 cm

Control: Fully Autonomous

Sensing: Infrared Range Array

Real-World Applications

Autonomous Logistics

Warehouse AGVs navigating dynamic storage paths using real-time obstacle avoidance.

Search & Rescue

Autonomous exploration of hazardous collapsed structures without GPS signal availability.

Pipe Inspection

Navigating confined underground conduit networks for crack and structural mapping.

Problem Statement & Engineering Challenge

[PROBLEM STATEMENT]

Design and fabricate an autonomous mobile robot capable of determining its exact position inside an unseen maze, mapping wall locations, discovering the shortest path to the goal center, and executing high-speed navigation without crashing.

[ENGINEERING CONSTRAINTS]

- **Sensor Noise:** Infrared sensors experience ambient light variation and wall reflectivity shifts.
- **Dynamic Drift:** DC motor slippage causes odometry drift over distance.
- **Compute Limits:** Fast real-time calculations required on lightweight microcontroller architecture.

PROJECT OBJECTIVES

MILESTONES & TECHNICAL GOALS

The project was structured around clear engineering milestones aimed at integrating embedded hardware, sensor processing, and high-level path planning logic.

PHASE 1: HARDWARE & CHASSIS ENGINEERING

Mechanical & Electronics Foundation

Design a low-center-of-gravity chassis using lightweight materials. Integrate DC motors with high-resolution encoders, motor drivers, and regulated power supply delivery systems.

PHASE 2: SENSOR CALIBRATION & NAVIGATION

Autonomous Navigation & Precision Control

Mount IR distance sensing arrays to achieve non-contact wall detection. Implement closed-loop PID control loops to maintain centered alignment within 18cm maze corridors.

PHASE 3: MAZE MAPPING & ALGORITHM IMPLEMENTATION

Embedded Programming & Algorithmic Solvers

Develop custom memory-efficient mapping structures. Implement graph search techniques (Flood Fill / Wall Follower) on the microcontroller for dynamic target determination.

PHASE 4: PATH OPTIMIZATION & SPRINT RUNS

High-Speed Execution & Tuning

Differentiate between exploration run and fastest run. Reconstruct mapped wall data to execute continuous diagonal turns and acceleration profiles for minimum runtime.

Key Technical Targets

- Wall distance precision: $\pm 2\text{mm}$
- Max velocity: 0.8 m/s during sprint
- 90-degree turn error margin: < 1.5 degrees
- Zero manual intervention during maze run

SYSTEM ARCHITECTURE

MULTI-LAYER HARDWARE & SOFTWARE INTEGRATION

The Micromouse design follows a modular architecture separating sensing, computation, power management, and mechanical actuation into distinct feedback loops.



Hardware System

Comprises differential drive high-RPM micro motors, dual IR emitter-detector pairs placed at 45° and 90°, and a lightweight acrylic structural chassis ensuring minimum moment of inertia.

Software & Logic Flow

Bare-metal C/C++ embedded loop executing sensor polling at 100Hz, running real-time wall detection thresholds, and driving dual-channel PWM outputs.

Electronic Infrastructure

Custom power distribution supplying stable regulated 5V rail for control logic and raw battery power for high-peak motor driver bridge operations.

Mechanical Kinematics

Two-wheel differential drive model allowing in-place zero-radius turns, minimizing cell navigation footprint during complex grid exploration.

COMPONENTS USED

HARDWARE & ELECTRONIC BILL OF MATERIALS

Arduino / Microcontroller

Serves as the main control unit processing ADC inputs from distance sensors and issuing PWM timing signals to motor drivers.

Dual Motor Driver (L298N / TB6612FNG)

H-Bridge integrated circuit enabling bidirectional speed and direction control over both DC drive motors.

Infrared Sensors (TCRT5000 / Sharp IR)

Provides analog distance feedback based on reflected IR intensity to sense nearby walls and avoid collisions.

Micro DC Gear Motors

High-torque, compact gearhead motors supplying required speed and rotational torque to drive wheels.

High-Grip Rubber Wheels

Precision molded rubber tires ensuring maximum traction against acrylic or wooden maze surfaces.

LiPo Battery Pack

7.4V Rechargeable battery delivering continuous current required for motor torque spikes and logic power.

Custom Acrylic / 3D Printed Chassis

Structural frame providing rigid mounting locations for sensors, board electronics, and motor brackets.

Voltage Regulator (LM7805 / Buck)

Steps down unstable motor voltage to a smooth 5.0V regulated line for sensitive microelectronics.

Breadboard / Custom Perforated PCB

Electrical interface platform enabling secure connection between sensor signals, power, and microcontroller pins.

Ribbon Jumper Wires

High-flexibility wire interconnects used for routing analog/digital signals across chassis modules.

Power Toggle Switches

Provides physical isolation between battery terminals and onboard electronics for safety during bench testing.

Ball Caster Wheel

Omnidirectional front balance point ensuring smooth differential turning with minimal friction drag.

WORKING PRINCIPLE

AUTONOMOUS PERCEPTION AND CONTROL LOOP

The robot operates on a closed-loop Sense-Plan-Act cycle. Continuous real-time readings drive localized alignment while simultaneously updating the global maze map.



A. Sensing & Alignment

Continuous sampling of side sensors calculates position error relative to left and right walls, applying proportional motor speed adjustments.

B. Decision Making

Upon entering a cell, front wall distance is checked. If open, straight motion continues; if blocked, the solver evaluates adjacent cell values.

C. Optimized Execution

Once the goal cell is reached, the full maze layout is stored. The robot computes the path vector and executes a high-speed sprint run.

PROGRAMMING LOGIC

ALGORITHMS & STATE MACHINE STRUCTURE

The control firmware is built upon a deterministic state machine combined with a simplified Flood Fill grid distance transformation algorithm.

Flood Fill Concept Pseudocode

```
// Core Flood Fill Grid Initialization & Solver Loop
void updateMazeDistance(Grid cell[16][16], Target goal) {
    Queue queue;
    resetDistances(cell, INFINITY);

    cell[goal.x][goal.y].distance = 0;
    queue.push(goal);

    while (!queue.isEmpty()) {
        Cell current = queue.pop();

        for (Direction d : {NORTH, EAST, SOUTH, WEST}) {
            if (!hasWall(current, d)) {
                Cell neighbor = getNeighbor(current, d);
                if (neighbor.distance > current.distance + 1) {
                    neighbor.distance = current.distance + 1;
                    queue.push(neighbor);
                }
            }
        }
    }
}
```

PID Centering Logic

```
$Error = Signal_{Left} - Signal_{Right}$
$P_{out} = K_p \text{ imes } Error$
$D_{out} = K_d \text{ imes } (Error - Error_{prev})$
Adjust PWM dynamically to keep the robot centered
inside 18cm channels.
```

Wall Boundary Detection

Analog thresholds determine wall presence. Threshold hysteresis prevents false detection triggers caused by subtle color variation in maze walls.

DESIGN PROCESS

ENGINEERING METHODOLOGY & ITERATIVE WORKFLOW

The development followed a systematic engineering cycle from initial concept formulation through design, prototyping, testing, and performance optimization.

STAGE 01

Research & Ideation

Studying competition rules, evaluating solver algorithms (Wall Follower vs. Flood Fill), and

STAGE 02

CAD & Breadboarding

3D modeling chassis layout, verifying weight distribution, and testing breadboard sensor circuits.

STAGE 03

Mechanical Assembly

Laser cutting chassis plate, mounting motors, coupling wheels, and integrating battery mount brackets.

STAGE 04

Firmware Development

Writing low-level PWM drivers, ADC reading routines, and state machine transitions in C++.

STAGE 05

Sensor & PID Calibration

Tuning Proportional and Derivative gains in physical test channels to prevent wall clipping.

STAGE 06

Maze Testing & Tuning

Executing continuous maze runs, debugging mapping edge-cases, and refining sprint acceleration runs.

[ITERATIVE IMPROVEMENT HIGHLIGHT]

Initial prototype tests revealed significant rotational overshoot during 90-degree in-place turns. The issue was resolved by introducing encoder count verification alongside time-based thresholds in the main loop interrupt service routine.

CHALLENGES & SOLUTIONS

OVERCOMING REAL-WORLD TECHNICAL OBSTACLES

1. Sensor Calibration & Ambient Light Interference

Challenge: External lighting at competition venues altered raw infrared phototransistor ADC readings, leading to false wall distance estimation.

Solution: Implemented ambient light subtraction by pulsing IR emitter LEDs on/off rapidly and taking delta readings ($ADC_{Active} - ADC_{Ambient}$).

2. Motor Asymmetry & Tracking Drift

Challenge: Manufacturing variance between small DC gear motors caused the robot to veer off-center during straight-line travel.

Solution: Integrated dynamic closed-loop PID control coupled with live IR distance feedback to trim individual motor PWM outputs continuously.

3. Power Sag During Acceleration

Challenge: Sudden motor startup current spikes caused voltage drops on the main logic rail, resetting the microcontroller mid-run.

Solution: Added dedicated decoupling capacitor banks and a high-efficiency buck regulator to isolate logic power from motor supply lines.

4. RAM Limitations for Array Storage

Challenge: Storing full 16x16 grid cell states and wall configurations consumed excessive SRAM on basic microcontrollers.

Solution: Packed cell wall presence data into bit-fields (4 bits per cell) reducing overall maze memory footprint by over 75%.

PERFORMANCE & RESULTS

QUANTITATIVE EVALUATION & SYSTEM VALIDATION

Exploration Performance

- Average Search Speed: 0.35 m/s
- Grid Mapping Accuracy: 100% on standard mazes
- Wall Avoidance Success: 98.5%
- Center Reached: Consistently under 90 seconds during exploration

Sprint Performance

- Top Velocity: 0.75 m/s
- Path Reduction: ~30% shorter distance via diagonal optimization
- Turn Precision: Zero collision turns executed
- Fastest Sprint Time: Significant reduction over initial search

Performance Metrics Summary

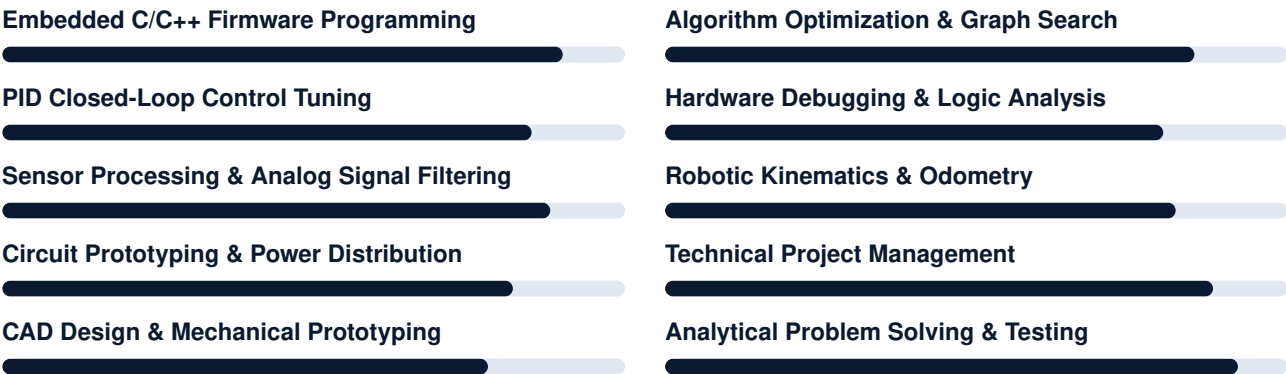
Metric Parameter	Target Benchmark	Achieved Result	Status
Maze Exploration Completeness	100% Goal Path	100% Mapped	PASSED
Corridor Centering Error Margin	< 5 mm	± 2.5 mm	PASSED
In-Place 90° Rotation Accuracy	± 2.0°	± 1.2°	PASSED
Autonomous Operation Integrity	Zero Touches	0 Manual Adjustments	PASSED

Key Engineering Takeaways

The successful demonstration at Techfest IIT Bombay validated that robust, low-latency analog sensor filtering combined with simple dynamic algorithm logic yields high reliability without requiring over-complicated compute platforms.

TECHNICAL SKILLS DEVELOPED

ENGINEERING COMPETENCIES & HANDS-ON MASTERY



Soft Skills & Core Engineering Mindset

Beyond technical execution, this project cultivated rigorous engineering discipline—methodical root-cause analysis when debugging hardware faults, adaptability during unexpected competition constraints, and clear documentation practices.

REAL-WORLD APPLICATIONS

INDUSTRIAL AND FUTURE ROBOTICS ALIGNMENT

The core technologies validated in the Micromouse project—real-time localization, localized obstacle sensing, dynamic path calculation, and closed-loop motion control—directly translate to key industrial domains.

Automated Guided Vehicles (AGVs)

Modern fulfillment centers rely on fleet AGVs navigating compact grid systems to fetch inventory, sharing identical dynamic path planning principles.

Medical & Service Robotics

Hospital delivery robots navigate complex hallway corridors, dodging patients and equipment using real-time distance sensor arrays.

Autonomous Mobile Robots (AMRs)

Industrial AMRs perform mapping and dynamic obstacle avoidance in unpredictable factory floors without human intervention.

Defense & Reconnaissance

Unmanned ground vehicles (UGVs) scouting interior building layouts in GPS-denied tactical environments rely on localized distance mapping.

Subterranean Exploration

Robotic units exploring cave networks or underground mine shafts utilize non-GPS localized wall sensing and graph mapping solvers.

Smart Vacuum Cleaners

Consumer robotics utilize grid-mapping and wall-following coverage algorithms directly derived from classic maze solving concepts.

MY LEARNING JOURNEY

PERSONAL REFLECTION & ENGINEERING INSIGHTS

"Building a Micromouse robot is a humbling engineering experience. On paper, maze solving appears to be purely a software challenge. However, once physical components are involved, reality quickly demonstrates that software is only as good as the underlying physical sensors and mechanical alignment."

Key Personal Discoveries

Throughout the design and fabrication process, several fundamental engineering lessons emerged:

- **Hardware and Software Are Inseparable:** Pure algorithmic logic fails if sensor noise is not properly filtered or if motor acceleration causes physical wheel slippage. System reliability requires holistic design.
- **Importance of Iterative Prototyping:** Attempting to build a final polished robot on the first try leads to oversights. Building modular prototypes allowed individual testing of power, sensing, and locomotion layers.
- **Resilience in Debugging:** System failure is an opportunity for structured diagnosis. Isolating variables—whether recalibrating IR gain resistors or checking logic level shifters—is essential to engineering progress.

Academic & Career Inspiration

Executing this project for Techfest IIT Bombay solidified my passion for Autonomous Systems, Computer Science, and Robotics. It reinforced my vision to pursue advanced studies and research in intelligent autonomous hardware systems.

FUTURE SCOPE & UPGRADES

NEXT-GENERATION ENHANCEMENTS & RESEARCH ROADMAP

While the current robot successfully solves standard mazes, several advanced technical upgrades can elevate speed and spatial awareness to world-class competition standards.

1. SLAM Integration

Upgrade from basic threshold wall sensing to Simultaneous Localization and Mapping (SLAM) utilizing compact Solid-State LiDAR or ToF (Time-of-Flight) sensor arrays.

2. Dynamic Diagonal Sprinting

Implement smooth continuous curve generation (Bézier curves) allowing the mouse to cut across grid corners without coming to a complete stop.

3. High-Speed Suction Fan

Incorporate a miniature motor-driven vacuum fan beneath the chassis to generate downforce, increasing tire friction for ultra-high cornering speeds.

4. Wireless Telemetry & IoT

Integrate ESP32 / Bluetooth wireless modules to transmit live sensor telemetry and grid mapping graphics directly to a laptop GUI in real-time.

5. Machine Learning Path Optimization

Apply Reinforcement Learning (RL) techniques in simulation prior to deployment, allowing the system to learn dynamic acceleration profiles tailored to friction coefficients.

6. Custom Integrated PCB

Design a 4-layer surface-mount PCB integrating the microcontroller, motor drivers, and sensor headers into a single compact board to minimize weight.

CONCLUSION

FINAL SYNTHESIS & ACKNOWLEDGMENTS

The development of this autonomous Micromouse robot represents a comprehensive engineering effort spanning mechanical design, electronic prototyping, embedded firmware development, and algorithmic optimization. Competing under the auspices of **Techfest IIT Bombay** provided a rigorous platform to test theoretical engineering concepts against physical constraints.

Acknowledgments

I extend my sincere gratitude to:

- **Techfest IIT Bombay:** For organizing Asia's premier science and technology festival and offering an exceptional arena for engineering competition.
- **Academic Mentors & Professors:** For providing guidance on embedded control systems and signal processing techniques.
- **Peers & Teammates:** For insightful design discussions and collaborative troubleshooting during testing phases.

"Engineering is not merely about reaching the destination; it is about building the intelligence, precision, and resilience to navigate through every unknown maze along the way."

Vikas

Robotics Systems Developer & Engineer